

QWriter: Technology-Enhanced Alphabet Acquisition based on Reinforcement Learning

Aidar Shakerimov
Department of Robotics and
Mechatronics
School of Engineering and Digital
Sciences
Nazarbayev University
Astana, Kazakhstan

Nurziya Oralbayeva
Graduate School of Education
Nazarbayev University
Astana, Kazakhstan

Shamil Sarmonov
Department of Robotics and
Mechatronics
School of Engineering and Digital
Sciences
Nazarbayev University
Astana, Kazakhstan

Aida Zhanatkyzy
Department of Robotics and
Mechatronics
School of Engineering and Digital
Sciences
Nazarbayev University
Astana, Kazakhstan

Aida Amirova
Graduate School of Education
Nazarbayev University
Astana, Kazakhstan

Zhansaula Telisheva
Department of Robotics and
Mechatronics
School of Engineering and Digital
Sciences
Nazarbayev University
Astana, Kazakhstan

Arna Aimysheva
Department of Robotics and
Mechatronics
School of Engineering and Digital
Sciences
Nazarbayev University
Astana, Kazakhstan

Anara Sandygulova*
Department of Robotics and
Mechatronics
School of Engineering and Digital
Sciences
Nazarbayev University
Astana, Kazakhstan

ABSTRACT

In Kazakhstan, the ongoing Cyrillic-to-Latin alphabet shift raises challenges for early literacy development and acquisition in the Kazakh language. This paper proposes the QWriter system to help young children learn the Latin-based Kazakh alphabet and its handwriting. The system consists of a humanoid robot NAO, a tablet with a stylus, and a Reinforcement Learning (RL) agent that learns a child's mistakes and progress to maximize alphabet learning in the shortest period of time by adapting the order of practice words according to the child's mistakes. To evaluate the effectiveness of the QWriter system, we conducted a between-subject design experiment with 59 Kazakh children aged 6-8 years old and compared their learning performance with a human tutor and the CoWriting Kazakh robot system. The results did not support our assumption, we found that the proposed system received significantly higher likability scores than the baseline human tutor.

CCS CONCEPTS

- **Human-centered computing** → **Empirical studies in HCI**;
Field studies; • **Computer systems organization** → **Robotics**;
• **Applied computing**;

KEYWORDS

child-robot interaction, robot-assisted language learning, Kazakh, reinforcement learning

ACM Reference Format:

Aidar Shakerimov, Shamil Sarmonov, Aida Amirova, Nurziya Oralbayeva, Aida Zhanatkyzy, Zhansaula Telisheva, Arna Aimysheva, and Anara Sandygulova*. 2023. QWriter: Technology-Enhanced Alphabet Acquisition based on Reinforcement Learning. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544549.3585611>

1 INTRODUCTION

Central to this work is Reinforcement Learning (RL) which has gained traction in human-computer interaction (HCI) [den Hengst et al. 2020]. It is a framework for decision-making processes in which an agent is capable of choosing actions based on the interaction with its environment to learn an optimal behavior [Akalın and Loutfi 2021; Sutton and Barto 2018]. Evidence from HCI research has shown that RL can be effectively applied in computer-assisted language learning (CALL) [Su et al. 2013], mobile and web-based language learning applications [Heil et al. 2016], educational robotic

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '23, April 23–28, 2023, Hamburg, Germany
© 2023 Copyright held by the owner/author(s).
ACM ISBN 978-1-4503-9422-2/23/04.
<https://doi.org/10.1145/3544549.3585611>

systems [Gordon et al. 2016], dialogue systems [Merdivan et al. 2019], with significant impact on learning outcomes of different age groups through adjustment of activities based on usage patterns. Some earlier research [Chen et al. 2020; Gordon et al. 2016; Leyzberg et al. 2014] have shown the effective application of the RL system in language learning and other educational tasks in human-robot interaction (HRI), one of the subbranches of HCI. As mentioned in previous HRI literature (e.g., Tapus et al. [2007]), creating an adaptive robot system for real-world use is one of the grand challenges in the field of social robotics.

In this work, we focus on the alphabet switch from Kazakh Cyrillic to Kazakh Latin which sets a unique research context. It also brings critical challenges for early literacy acquisition and development in children who have to learn to write in the two alphabets at the same time. These completely different writing systems may cause confusion among young children to stay motivated and use both alphabets. Specifically, the children might lose interest and motivation to write and read in the Kazakh Latin alphabet due to the lack of educational and entertaining content that will be available in the new script. Our work responds to these caveats and aims to design and create an effective and engaging technology-supported learning system for the acquisition of the Kazakh Latin alphabet. We propose the QWriter system which integrates RL-based learning into the decision-making of alphabet learning application to provide an adaptive learning experience while maximizing learning gains in the shortest period of time. To evaluate its effectiveness, we conducted a user study with children from Grade 1 in a public school. We attempt to investigate whether the proposed QWriter system with adaptive capabilities can be comparable to the non-adaptive CoWriter robot and a human tutor in alphabet practice. In the current study, the human tutor condition offers a baseline for what we see as a technology-enhanced way of learning in the modern world (i.e., a teacher using a tablet). The use of CoWriting Kazakh system in addition to QWriter and human tutor can help us to see how robot roles and behaviors may impact learning and teaching in the HRI context.

This research makes three important contributions. First, it proposes a technology-enhanced environment for the acquisition of the alphabet and handwriting practice. The use of an easy-to-use tablet and physically embodied robots contributes to multi-modal learning, addressing real-world challenges such as early literacy development amidst the alphabet switchover. A robot using a tablet closely resembles a human teacher with a tablet while teaching students in a real classroom [Hu et al. 2022]. Second, our research provides empirical insights to the real-world application of the RL-based agent. Most RL-driven works are limited to simulated datasets and show little utility in unrestrained everyday settings [Serban et al. 2018]. Hence we expect that applying the QWriter system *in the wild* can evaluate the real-world utility. Thus far, RL-based robotic systems and human tutors are rarely compared to on their effectiveness as a tutor. We should state clearly that the comparison is not to test if the adaptive robot can replace a human tutor, but explore its potential compared to the teacher providing natural reinforcement in the same learning setting. We know that assistive learning systems are already present in today's classrooms, and therefore the use of social robots in educational settings is just a matter of time. Finally, given that human tutoring

in large classrooms is a labor- and time-intensive undertaking, this work suggests some promising prospects for teachers to navigate technology-embedded classrooms. When teaching learning skills like handwriting, it is vital to provide affordable learning opportunities via automated solutions that come with their own reward mechanisms.

2 BACKGROUND LITERATURE

2.1 Applications of Reinforcement Learning in Language Education

As educational technologies have unfolded in recent years, the application of Reinforcement Learning algorithms has gained its avenue in scholarly inquiries, being implemented on various platforms (e.g., embodied conversational agents, chatbots, virtual avatars, social robots). In the social HRI works, personalized tutoring has been commonly addressed from the perspective of affective policy learning, for instance, affect-focused states such as engagement have been explored to build reward signals in RL-based frameworks to choose motivational strategies or supportive behaviors tailored to each learner [Gao et al. 2017]. RL-based frameworks also focus on various social behaviors to attain interactional goals in task-oriented HRI [Akalin and Loutfi 2021]. A few significant studies have explored the effectiveness of using RL algorithms in an engaging learning scenario with young children and a social robot. As an example, Gordon et al. [2016] compared the effect of personalized and non-personalized affective feedback in language learning with a social robot and a tablet for children. In doing so, they implemented the affective RL algorithm for long-term personalization. They found that the affective RL-based robot can substantially increase a long-term positive valence as compared to non-personalized interactions. Further evidence for the adaptive RL-based robot's benefits is presented in Chen et al. [2020], who utilized RL to construct an adaptive role-switching model (ARS) for a reciprocal peer robot in a vocabulary learning scenario. Their results showed that children who interacted with the personalized peer robot, which embodied both tutor and tutee roles through the ARS model, achieved the highest performance and word learning scores. RL has recently been employed in the alphabet learning context in a study by Oralbayeva et al. [2022], where they compared the effectiveness of using RL-based and naive chatbots for learning the Kazakh Latin alphabet among university students. Although no statistically significant differences were found in learning gains, the RL-based chatbot had higher likeability and intelligence scores due to its adaptive Q-Learning algorithm as rated by the students.

However, RL has several limitations linked to external rewards for learning and task accomplishment [Qureshi et al. 2018]. In HRI, the lack of direct rewards usually makes learning challenging and requires socially-inspired rewards based on speech and behavioral cues [Ferreira and Lefèvre 2015]. Reflecting on these limitations, we test the extent to which the proposed QWriter system can be effective in providing adaptive learning support as well as in stimulating cognitive gains in primary-age children.

2.2 The CoWriter project

Handwriting is a complex language skill and involves attention, perceptual, linguistic and fine motor skills [Bara and Gentaz 2011;

Gargot et al. 2020; Santangelo and Graham 2015]. It forms the foundation of language acquisition and affects an individual's learning progress. Hence handwriting development has been recognized in the HRI community with a focus on robot-assisted language learning. The CoWriter project and its active follow-up iReCHeCk explore how a robot can help children with learning disabilities in the acquisition of handwriting, presenting a learning approach - learning by teaching [Hood et al. 2015]. This approach encourages social learning in which a child takes on a role to "teach" another and then learns as much as a peer. By applying the interactive teaching approach with computer-based agents, this approach might bring motivational, meta-cognitive, and learning benefits [Jacq et al. 2016; Rohrbeck et al. 2003]. Jacq et al. [2016] tested the hypothesis that if the CoWriter is adapted to the learning environment, children could improve their handwriting experiences. The results show that the successful implementation was achieved due to self-correction on a child's part, leading to better handwriting. The project goals and its setup were later applied to a completely new context of learning an alphabet through biscriptal practices within the CoWriting Kazakh project [Sandygulova et al. 2019, 2020; Zhaxenova et al. 2020]. In Sandygulova et al. [2020], CoWriting Kazakh was successfully implemented in a new script-switching task. An English-speaking peer robot could facilitate children's learning experience and thus provide motivational support for alphabet learning. Later, Zhaxenova et al. [2020] compared three learning aids, a teacher, a robot with a tablet, and a tablet only, to identify what kind of learning environment works well for children to acquire the Kazakh Latin alphabet. That study did not find any significant differences in learning gains between robot and human teacher conditions, meaning more or less the same learning outcomes. The children rated the robot as the most favorite learning aid. The use of the CoWriter robotic system as a robot condition in the current study is thus motivated by the earlier works [Hood et al. 2015; Jacq et al. 2016; Zhaxenova et al. 2020] on early literacy acquisition and handwriting practices in the Kazakh language with a social robot.

3 QWRITER ROBOT

This section details the proposed QWriter robotic system, its hardware and software components.

3.1 Learning activity

Alphabet learning activity includes 12 Kazakh words selected out of 60 in the overall word list with all the letters of the Kazakh Latin alphabet at least once. The robot "writes" on a Wacom Cintiq Pro tablet one letter at a time. The child is asked then to write the letters by copying the robot's writing. If the child makes a mistake in the spelling or writing of the letters, the next practice word is selected accordingly. For example, if the word "saalem" is written incorrectly by the child, the robot offers corrective feedback and writes the incorrectly written letter again. The following word is then selected according to this mistake so that the child has an opportunity to practice this letter again. The system does not repeat words.

3.2 Hardware

In this work, we employ a social robot NAO developed by SoftBank Robotics. It is the most popular social robot with an autonomous

and programmable architecture. Its common application areas include education and healthcare [Amirova et al. 2021]. Young users appreciate its non-threatening and human-like appearance. On various parts of its body, it has seven touch sensors and 25 degrees of freedom. The child voice on the robot was a custom-made Kazakh speech synthesis. The robot is programmed to produce verbal and non-verbal social cues, such as hand gestures, body behaviors and speech utterances for a 30-minute autonomous activity. Another component of the system includes the Wacom Cintiq Pro tablet. The tablet serves as the second monitor when connected to a laptop. It has a stylus with an 8.192" of pressure sensitivity and tilting recognition. There were eyeglasses to gather eye-tracking data.

3.3 Reinforcement Learning

We propose the RL-based QWriter system (see Figure 1) which consists of an agent, environment and user interface. The agent presents an RL algorithm to learn action selection by interacting with the environment. The environment plays an intermediate role to connect the agent with a user by collecting information about the user's learning and mistakes. It also sends this knowledge to the agent, receives an action and finally uses it in interaction with the user.

The environment performs the writing of words. Within the environment, a set of variables is considered to be useful in the learning process. It contains several lists: a list of unexplored common Latin letters, a list of unexplored specific Kazakh letters, a list of problematic common Latin letters and a list of problematic Kazakh-specific letters (i.e. the child writes it with error), a set of available words, and the currently assigned word.

The state of the agent is developed from observations of letters and their quantity in the above-mentioned lists of unexplored and problematic letters. However, to keep the state-space small enough for faster exploration, we quantize the number of items in the lists as follows:

- The number of unexplored common Latin letters is split into three groups: a) zero letters, b) more than zero but less than six, c) more than six letters in the list;
- The number of unexplored Kazakh-specific letters is organized into three groups: a) zero letters, b) more than zero but less than six, c) more than six letters in the list;
- The number of problematic common Latin letters is grouped into three categories: a) zero letters, b) more than zero but less than six, c) more than six letters in the list;
- The number of problematic Kazakh-specific letters is listed in three groups: a) zero letters, b) more than zero but less than six, c) more than six letters in the list.

The state then is a tuple of four elements including the group index of the number of unexplored common Latin letters (L), the group index of the number of unexplored specific Kazakh letters (K), the group index of the number of problematic common Latin letters (PL), and the group index of the number of problematic specific Kazakh letters (PK).

3.3.1 Interaction with user. The agent chooses a word from the pool of words and imitates the handwriting letter by letter on the tablet. The robot then asks the child to repeat the writing of the chosen word using the stylus. The letters included in the word are

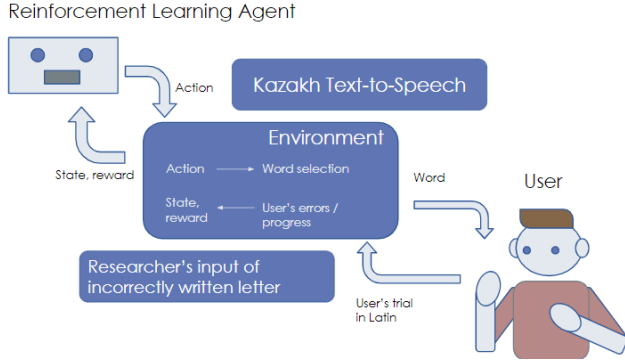


Figure 1: The environment interacts with an agent by providing observations and rewards and getting selected action. The interaction with a user is performed by displaying the selected word to a user and recording the mistakes as well as the duration of the interaction.

extracted from the list of unexplored letters. While recognizing the user's spelling, the environment captures mistakes. The resultant state is returned to the agent.

The agent chooses an action or does a random action based on epsilon-greedy policy (i.e. with a probability ϵ it takes a random action instead of the best learned one). The options for actions are below:

- (1) Select a word with unexplored common letters (if the corresponding list is not empty),
- (2) Select a word with unexplored Kazakh-specific letters (if the corresponding list is not empty),
- (3) Select a word with a common problematic letter (if the corresponding list is not empty),
- (4) Select a word with a problematic Kazakh-specific letter (if the corresponding list is not empty).

The environment shows a new word to the user in agreement with the given action selection and identifies the child's errors. The obtained data from the user interaction is returned as a new state to the agent. At the end of 12 iterations the system provides a reward by calculating the number of encountered challenges (both solved and unsolved errors), the number of unsolved errors (if a child kept writing a certain letter with an error) and the number of unexplored letters, where the number of unsolved problematic letters $N_{unsolved}$ is the sum of unsolved common Latin letters and unsolved Kazakh-specific letters multiplied by the factor of 3 (so that specific Kazakh letters obtain a higher priority):

The number of unexplored letters $N_{unexplored}$ is the sum of unexplored common Latin letters and unexplored specific Kazakh letters multiplied by the factor of 3 (so that specific Kazakh letters obtain a higher priority):

3.3.2 Reinforcement learning agent. To achieve an optimal behavior for increasing the reward we use the agent with a Q-learning algorithm [Watkins and Dayan 1992]. It computes and stores action

values for each state-action pair that it found. The action value is an estimated optimality of taking a specific action when the specific state is given. They are computed using the next formulas:

$$Q^{new}(s_t, a_t) = Q^{old}(s_t, a_t) + \alpha * (temporal\ difference) \quad (1)$$

where $Q^{new}(s_t, a_t)$ is a new value state-action pair (s_t, a_t) , $Q^{old}(s_t, a_t)$ is its old value, α is a learning rate (the factor describing the significance of a newer experience over an older experience), and temporal difference is given in the following formula:

$$temporal\ difference = r_t + \gamma * \max(Q(s_{t+1}, a)) - Q^{old}(s_t, a_t) \quad (2)$$

where r_t is a reward, $\max(Q(s_{t+1}, a))$ is an estimated maximal value for the next state, and γ is a discount factor which is multiplied to the estimated maximal value for the next state making recent rewards worthier than previous (we use $\gamma = 0.99$).

4 EXPERIMENT

4.1 Method

We conducted a between-subject design experiment in a primary school. The participants interacted with either a QWriter system, a CoWriter system or a human tutor for 30-40 minutes. Each condition included pair work. We carefully counterbalanced children by their age and gender with otherwise random assignment of child pairs to one of the three conditions.

4.2 Recruitment

The research was approved by The Institutional Research Ethics Committee at Nazarbayev University. Forms for assent and consent were distributed to children in front of their teachers. The aim of the study and the procedures were briefly explained to children in the presence of their teachers. Teachers collected the forms for us once the children returned the forms approved and signed by their parents.

4.3 Participants

A total of 66 children aged 6-8 years old participated in the experiment. Due to missing data of 7 children, the final analysis included 59 children (32 males and 27 females). At the time of the experiment, the children were attending the last school term. Their native language and language of instruction was Kazakh. They spent approximately nine months writing in Kazakh Cyrillic and had the same amount of writing experience in the English alphabet.

4.4 Hypotheses and Conditions

Based on our background literature and current evidence on RAAL, we attempt to validate the following hypotheses:

- The QWriter will be effectively implemented in an alphabet acquisition environment, bringing positive learning gains.
- Children will learn a similar number of letters when learning from the QWriter robot and the human tutor.
- Children interacting with the QWriter robot will learn more letters than those with the CoWriter robot.
- Children interacting with the CoWriter robot will gain better likability scores compared to the other two conditions.



Figure 2: Experimental conditions: (a) Human tutor, (b) QWriter and CoWriter robots

In this study, children interacted with one of the following three learning partners as shown in Figure 2:

QWriter robot (N=26): the order of 12 words is selected from the pool of 60 words based on each child’s mistakes. At this stage, the mistakes are entered by the researcher using keyboard on a laptop that is connected to the Wacom tablet. Along with the corrective feedback, the robot also provided auditory feedback to emphasize whether the child spelled the words correctly or not (e.g., “You got it right”). A custom-made Kazakh Text-to-Speech (TTS) synthesis was used to train on a child’s voice data. The system was semi-autonomous, which means the agent selected the words by itself, while the human tutor entered an incorrect letter into the system.

CoWriter robot (N=14): the robot takes the role of a peer who speaks English as a native language. The robot asks children for help to learn Kazakh while mentioning that the robot knows only Latin script. Therefore, the child is asked to translate the words (e.g., How do you say “hello” in Kazakh) and write them using Latin-based Kazakh for the robot to read them (“Could you write it using Latin script so that I can read it?”). The robot then asks children to read the word when they finish writing and repeats it after the children, with some foreign accent by saying “Let me also read it. It is a ...”. Here, the robot is fully autonomous. Then, the robot emulates writing in the air and gesticulates with its right hand. The CoWriter robot executed a few of the built-in upper body behaviors, arm gestures, and speech generated by the English TTS engine.

Human Tutor (N=19): the teacher writes a word on iPad using its stylus and asks the child to repeat it. This is done for 12 words, the order of which is fixed for each child. The teacher also provided feedback and corrected mistakes based on each child’s individual mistakes (e.g., “Let me show you how to write this letter”). The human tutor also used Kazakh for communication.

4.5 Procedure

The experiment procedure included a pre-test, a learning activity, and a post-test. First, two children were invited out of class and accompanied to the room with the robot by two researchers. The researchers tried to maintain a warm-up conversation with the kids while walking. Upon entering the room, children were asked to complete a pre-test to assess their knowledge of the Kazakh Latin

alphabet. Each child did the test on a separate table. After that, another researcher conducted a survey asking demographic questions such as age, linguistic background, and academic standing. Children were randomly assigned to take a seat at the table with one of the robots or with the human tutor. After the intervention, children were asked to complete a post-test in order to assess their learning gains. A semi-structured interview was then conducted with each child by one of the researchers to collect their perceptions of the interaction and the agent (a robot or a human). The researchers called for the next participants as soon as the children completed the task and returned to their classroom.

4.6 Measures

Learning Gain (LG) is computed from the difference between a pre-test and a post-test that children filled in before and after the intervention with the robot/human. A child was given a piece of paper with a list in two columns: one filled in with Kazakh Cyrillic letters, and the other column to be filled in with Kazakh Latin letters.

The Funometer scale [Risden et al. 1997] is a vertical scale, which resembles a thermometer, and which the children would use in a similar way, with a vertical bar representing the amount of fun [Markopoulos et al. 2008]. It allows us to record how much a child liked the interaction from 1 to 10 before and after the intervention. This would allow us to evaluate the most liked conditions.

5 RESULTS

We conducted a series of one-way and two-way ANOVA tests in order to see if there are any significant differences between three conditions as regards children’s learning gains and affective measures. A Shapiro-Wilk test was used on dependent variables and within conditions in order to check if they were normally distributed. Since some scores were significantly non-normal, non-parametric tests were applied.

A Wilcoxon signed-rank test revealed that the QWriter condition did not obtain any significant changes in terms of the children’s learning gain ($Z = 117.5, p = 0.6523$). Mainly, the children who participated in the QWriter condition, had their knowledge of the Kazakh Latin alphabet only slightly improved with an average number of learned letters of 1.15 ± 1.41 (Max = 5, Min = 0). According to the result of a one-way ANOVA, there was a statistically significant difference between the three conditions: $F(2,56) = 9.153, p = 0.0004$.

A Tukey post-hoc test revealed that children learned significantly less letters with QWriter (1.15 ± 1.41) and CoWriter (1.29 ± 1.44) robots compared to Human Tutor (3.42 ± 2.57). It did not reveal any significant differences between the two robots. When analyzed by the Funometer scales, there were statistically significant results in one-way ANOVA tests by condition: $F(1,57) = 4.404$, $p = 0.0167$. A Tukey post-hoc test revealed that children in the QWriter robot (9.69 ± 0.618) and CoWriter (9.86 ± 0.363) conditions rated the sessions significantly higher on a Funometer scale than the human tutor condition (8.58 ± 2.39).

6 DISCUSSION

The study evidences that it is possible to apply the QWriter system with children to increase learning gains. Overall, the mean learning gain averaged one letter, ranging from 0 to 5. This result shows a slightly positive learning outcome and allows us to accept H1. In H2, we assumed the QWriter robot would be as good as the human tutor, considering its corrective feedback and immediate reward to maximize alphabet retention. There was a significant double increase in the number of learned letters among children that interacted with the human tutor. H2 is thus refuted. Indeed, a single-session study cannot reliably explain learning experiences in measurable ways. From this, it follows that conducting long-term interventions is key to determining the RL-based robot's benefits, and identifying the appropriate algorithm to support technology-driven adaptation. As per H3, the study found no significant difference in learning gains between the QWriter robot and CoWriter systems. Both robotic agents provided more or less equal learning support for the children, revealing a similar learning effect in the two conditions. However, the learning gains were marginally high with the peer-like CoWriter robot. Although this result refutes the H3, it can be observed that the peer-like CoWriter robot was the second most effective learning aid after the human tutor. In Zhexenova et al. [2020], the CoWriter robot brought a similar rate of learning gains as the human teacher. The coach-like QWriter robot emulated commanding behaviors such as explaining what to do next and correcting the mistakes after each word practice. This appears to discourage children to write letters as taught by the robot behaving like a "traditional teacher". In recent research [Chen et al. 2020], the peer-like RL-based agent brought the most benefit to children's learning compared to a tutor agent. We may consider designing and introducing the QWriter robot as a peer that is the same age and has the same social abilities. Finally, we accept H4 because the children's self-reports showed a significantly higher likeability toward the CoWriter robot. Unlike the QWriter, the CoWriter robot did not teach children directly and maintained a peer-like interaction with them. We believe its collaborative role could enhance likability scores. That being said, the tutor-like QWriter robot was rated as the second most favorite agent compared to the human tutor. Robots are generally more advantageous in enabling positive mood, which is often attributed to the newness of and excitement around their social presence [Stock-Homburg 2022]. However, when the novelty wears off, children may lose interest to interact with the robot and its long-term acceptance remains unexplored [Abendschein et al. 2022].

Overall results show that the proposed QWriter robot was found to be useful to teach the alphabet in such a short period of time. However, its effectiveness was not comparable to that of the human tutor. The two robotic systems were rated more likable than the human tutor, with the peer-like CoWriter robot scoring the highest likability. There are still many unanswered questions on the long-term effects of RL systems, and this study has some space to be improved in future research. There might occur different learning and affective outcomes over repeated interactions. The variety in how a human may respond spontaneously (in both human tutor and robot conditions) makes it challenging to reach better conclusions, which should be considered in future evaluations. Contrary to most studies maintaining one-to-one HRI [Amirova et al. 2021], the children in this study worked in pairs with the assigned agent in a triadic context, which may result in different levels of pair engagement; which is not the focus of the current work. There might be limitations in how we interpret data from pair work in just a single session. We will continue to refine and apply this triadic learning scenario by conducting a long-term study with older students from other primary grades. Future research should also investigate adaptive interventions tailored to an individual child's learning progress through the implementation of RL algorithms on dyadic vs. triadic HRI. The QWriter system may be introduced as an assistant to mediate teaching-learning contact between children and a teacher. This is derived from our results showing higher learning gains yet lower likeability scores for the human teacher. The presence of novel technologies may support and increase the efficiency of the human teacher as well as positive affective engagement due to the robot's presence. Our future evaluation will consider long-term interactions with the QWriter system and design the robot as a peer to a child or a teacher's assistant in tutoring sessions, whereby agents switch roles to learn from one another.

ACKNOWLEDGMENTS

The authors would like to extend their gratitude to the students who participated in the study. The study was funded by the Nazarbayev University Collaborative Research Program grants (091019CRP2107, 11022021CRP1502).

REFERENCES

- Bryan Abendschein, Autumn P. Edwards, and Chad Edwards. 2022. Novelty Experience in Prolonged Interaction: A Qualitative Study of Socially-Isolated College Students' In-Home Use of a Robot Companion Animal. *Frontiers in Robotics and AI* 9 (2022), 1-19 pages.
- Neziha Akalin and Amy Loutfi. 2021. Reinforcement learning approaches in social robotics. *Sensors* 21, 4 (2021), 1292.
- Aida Amirova, Nazerke Rakhymbayeva, Elmira Yadollahi, Anara Sandygulova, and Wafa Johal. 2021. 10 Years of Human-NAO Interaction Research: A Scoping Review. *Frontiers in Robotics and AI* 8 (2021), 744526. <https://doi.org/10.3389/frobt.2021.744526>
- Florence Bara and Edouard Gentaz. 2011. Haptics in teaching handwriting: The role of perceptual and visuo-motor skills. *Human movement science* 30 (08 2011), 745–59. <https://doi.org/10.1016/j.humov.2010.05.015>
- Huili Chen, Hae Won Park, and Cynthia Breazeal. 2020. Teaching and learning with children: Impact of reciprocal peer learning with a social robot on children's learning and emotive engagement. *Computers & Education* 150 (2020), 1-22 pages. <https://doi.org/10.1016/j.compedu.2020.103836>
- Floris den Hengst, Eoin Martino Grua, Ali el Hassouni, and Mark Hoogendoorn. 2020. Reinforcement learning for personalization: A systematic literature review. *Data Science* 3, 2 (2020), 107–147.
- Emmanuel Ferreira and Fabrice Lefèvre. 2015. Reinforcement-Learning Based Dialogue System for Human-Robot Interactions with Socially-Inspired Rewards. *Comput.*

- Speech Lang.* 34, 1 (nov 2015), 256–274.
- Alex Yuan Gao, Wolmet Barendregt, and Ginevra Castellano. 2017. Personalised human-robot co-adaptation in instructional settings using reinforcement learning. In *IVA Workshop on Persuasive Embodied Agents for Behavior Change: PEACH 2017*. TU Delft, Stockholm, Sweden, 1–10 pages.
- Thomas Gargot, Thibault Asselborn, Hugues Pellerin, Ingrid Zammouri, Salvatore M Anzalone, Laurence Casteran, Wafa Johal, Pierre Dillenbourg, David Cohen, and Caroline Jolly. 2020. Acquisition of handwriting in children with and without dysgraphia: A computational approach. *PLoS ONE* 15 (2020), 1–22 pages.
- Goren Gordon, Samuel Spaulding, Jacqueline Kory Westlund, Jin Joo Lee, Luke Plummer, Marayna Martinez, Madhurima Das, and Cynthia Breazeal. 2016. Affective Personalization of a Social Robot Tutor for Children's Second Language Skills. In *Proceedings of the Thirtieth AAAI Conference on Artificial Intelligence (AAAI'16, Vol. 30)*. AAAI Press, Phoenix, Arizona, 3951–3957.
- Catherine Regina Heil, Jason S Wu, Joey J Lee, and Torben Schmidt. 2016. A review of mobile language learning applications: Trends, challenges, and opportunities. *The EuroCALL Review* 24, 2 (2016), 32–50.
- Deanna Hood, Séverin Lemaignan, and Pierre Dillenbourg. 2015. The CoWriter Project: Teaching a Robot How to Write. In *The 10th ACM/IEEE International Conference on Human Robot Interaction (Portland, Oregon, USA) (HRI'15 Extended Abstracts)*. Association for Computing Machinery, New York, NY, USA, 269. <https://doi.org/10.1145/2701973.2702091>
- Chih-Chien Hu, Yu-Fen Yang, and Nian-Shing Chen. 2022. Human-robot interface design – the 'Robot with a Tablet' or 'Robot only', which one is better? *Behaviour & Information Technology* 1 (07 2022), 1–14. <https://doi.org/10.1080/0144929X.2022.2093271>
- Alexis Jacq, Séverin Lemaignan, Fernando Garcia, Pierre Dillenbourg, and Ana Paiva. 2016. Building successful long child-robot interactions in a learning context. In *2016 11th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. IEEE Press, Christchurch, New Zealand, 239–246. <https://doi.org/10.1109/HRI.2016.7451758>
- Daniel Leyzberg, Samuel Spaulding, and Brian Scassellati. 2014. Personalizing Robot Tutors to Individuals' Learning Differences. In *Proceedings of the 2014 ACM/IEEE International Conference on Human-Robot Interaction*. Association for Computing Machinery, New York, NY, USA, 423–430. <https://doi.org/10.1145/2559636.2559671>
- Panos Markopoulos, Janet C Read, Stuart MacFarlane, and Johanna Hoysniemi. 2008. *Evaluating Children's Interactive Products: Principles and Practices for Interaction Designers*. Morgan Kaufmann Publishers Inc., San Francisco, CA, USA.
- Erinc Merdivan, Deepika Singh, Sten Hanke, and Andreas Holzinger. 2019. Dialogue systems for intelligent human computer interactions. *Electronic Notes in Theoretical Computer Science* 343 (2019), 57–71.
- Nurziya Oralbayeva, Aidar Shakerimov, Shamil Sarmonov, Kanagat Kantoreyeva, Fatima Dadebayeva, Nuray Serkali, and Anara Sandygulova. 2022. K-Qbot: Language Learning Chatbot Based on Reinforcement Learning. In *HRI '22: Proceedings of the 2022 ACM/IEEE International Conference on Human-Robot Interaction (HRI '22)*. IEEE Press, Sapporo, Hokkaido, Japan, 963–967.
- A. H. Qureshi, Yutaka Nakamura, Yuichiro Yoshikawa, and Hiroshi Ishiguro. 2018. Intrinsically motivated reinforcement learning for human-robot interaction in the real-world. *Neural networks : the official journal of the International Neural Network Society* 107 (2018), 23–33. <https://doi.org/10.1016/j.neunet.2018.03.014>
- K Risdén, E Hanna, and A Kanerva. 1997. Dimensions of intrinsic motivation in children's favorite computer activities. *Society for Research in Child Development, Washington, DC* 1 (1997), 7.
- Cynthia A. Rohrbeck, Marika Ginsburg-Block, John W. Fantuzzo, and Traci Rinker Miller. 2003. Peer-assisted learning interventions with elementary school students: A meta-analytic review. *Journal of Educational Psychology* 95 (2003), 240–257.
- Anara Sandygulova, Anna CohenMiller, Nurziya Oralbayeva, Wafa Johal, Thibault Asselborn, and Pierre Dillenbourg. 2019. Cognitive Learning with a Robot: The Case of Script Acquisition. In *Software Engineering and Formal Methods: SEFM 2019 Collocated Workshops: CoSim-CPS, ASYDE, CIFMA, and FOCLASA, Oslo, Norway, September 16–20, 2019, Revised Selected Papers (Oslo, Norway)*. Springer-Verlag, Berlin, Heidelberg, 154–162. https://doi.org/10.1007/978-3-030-57506-9_12
- Anara Sandygulova, Wafa Johal, Zhanel Zhaxenova, Bolat Tleubayev, Aida Zhanatkyzy, Aizada Turarova, Zhansaule Telisheva, Anna CohenMiller, Thibault Asselborn, and Pierre Dillenbourg. 2020. CoWriting Kazakh: Learning a New Script with a Robot. In *Proceedings of the 2020 ACM/IEEE International Conference on Human-Robot Interaction*. IEEE, Cambridge, United Kingdom, 113–120.
- Tanya Santangelo and Steve Graham. 2015. A Comprehensive Meta-analysis of Handwriting Instruction. *Educational Psychology Review* 28 (2015). <https://doi.org/10.1007/s10648-015-9335-1>
- Julian Vlad Serban, Ryan Lowe, Peter Henderson, Laurent Charlin, and Joelle Pineau. 2018. A survey of available corpora for building data-driven dialogue systems. *Dialogue & Discourse* 9 (2018), 1–49 pages.
- Ruth Stock-Homburg. 2022. Survey of Emotions in Human-Robot Interactions: Perspectives from Robotic Psychology on 20 Years of Research. *Int. J. Soc. Robotics* 14 (2022), 389–411.
- Pei-hao Su, Yow-Bang Wang, Tien-han Yu, and Lin-shan Lee. 2013. A dialogue game framework with personalized training using reinforcement learning for computer-assisted language learning. In *2013 IEEE International Conference on Acoustics, Speech and Signal Processing*. IEEE, Vancouver, BC, Canada, 8213–8217.
- Richard S Sutton and Andrew G Barto. 2018. *Reinforcement learning: An introduction*. MIT press, Cambridge, Massachusetts.
- Adriana Tapus, Maja J. Mataric, and Brian Scassellati. 2007. Socially assistive robotics [Grand Challenges of Robotics]. *IEEE Robotics & Automation Magazine* 14, 1 (2007), 35–42. <https://doi.org/10.1109/MRA.2007.339605>
- Christopher JCH Watkins and Peter Dayan. 1992. Q-learning. *Machine learning* 8, 3–4 (1992), 279–292.
- Zhanel Zhaxenova, Aida Amirova, Manshuk Abdikarimova, Kuanyshe Kudaibergenov, Nurakhmet Baimakhan, Bolat Tleubayev, Thibault Asselborn, Wafa Johal, Pierre Dillenbourg, Anna CohenMiller, and Anara Sandygulova. 2020. A Comparison of Social Robot to Tablet and Teacher in a New Script Learning Context. *Frontiers in Robotics and AI* 7 (2020), 1–14 pages. <https://doi.org/10.3389/frobt.2020.00099>